

1. Write a Python program to print numbers from 1 to 10 using a for loop.

Code:

```
for i in range(1, 11):  
    print(i)
```

Input:

No input

Output:

```
1  
2  
3  
4  
5  
6  
7  
8  
9  
10
```

2. Write a Python program to print the multiplication table of a given number using a for loop.

Code:

```
n = int(input())  
for i in range(1, 11):  
    print(n, "x", i, "=", n * i)
```

Input:

5

Output:

```
5 x 1 = 5  
5 x 2 = 10  
5 x 3 = 15  
5 x 4 = 20  
5 x 5 = 25  
5 x 6 = 30  
5 x 7 = 35  
5 x 8 = 40  
5 x 9 = 45  
5 x 10 = 50
```

3. Write a Python program to find the sum of numbers from 1 to N.

Code:

```
n = int(input())  
sum = 0  
for i in range(1, n + 1):  
    sum = sum + i  
print(sum)
```

Input:

5

Output:

15

4. Write a Python program to find the first N Fibonacci numbers using a for loop.

Code:

```
n = int(input())  
a = 0  
b = 1  
for i in range(n):  
    print(a, end=" ")  
    a, b = b, a + b
```

Input:

7

Output:

0 1 1 2 3 5 8

5. Write a Python program to find the LCM of two numbers using a while loop.

Code:

```
a = int(input())
b = int(input())
lcm = max(a, b)
while lcm % a != 0 or lcm % b != 0:
    lcm = lcm + 1
print(lcm)
```

Input:

```
12
18
```

Output:

```
36
```

6. Write a Python program to generate Pascal's Triangle using nested loops.

Code:

```
n = int(input())
prev = []
for i in range(n):
    row = [1] * (i + 1)
    for j in range(1, i):
        row[j] = prev[j - 1] + prev[j]
    print(*row)
    prev = row
```

Input:

```
5
```

Output:

```
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
```

7. Write a Python program to count the number of occurrences of a character in a given string.

Code:

```
s = input()
ch = input()
count = 0
for i in s:
    if i == ch:
        count = count + 1
print(count)
```

Input:

```
hello world
l
```

Output:

```
3
```

8. Write a Python program to reverse a string using slicing.

Code:

```
s = input()
reverse = s[::-1]
print(reverse)
```

Input:

```
hello
```

Output:

```
olleh
```

9. Write a Python program to check if a given string is a palindrome or not.

Code:

```
s = input()
if s == s[::-1]:
    print("Palindrome")
else:
```

```
print("Not Palindrome")
```

Input:

madam

Output:

Palindrome

10. Write a Python program to capitalize the first letter of each word in a given string.

Code:

```
s = input()
print(s.title())
```

Input:

hello world python

Output:

Hello World Python

11. Write a Python program to find the length of the longest word in a given sentence.

Code:

```
s = input()
words = s.split()
longest = 0
for word in words:
    if len(word) > longest:
        longest = len(word)
print(longest)
```

Input:

Python is very interesting

Output:

11

12. Write a Python program to concatenate two strings without using the '+' operator.

Code:

```
s1 = input()
s2 = input()
print(s1, s2, sep="")
```

Input:

Hello
World

Output:

HelloWorld

13. Write a Python program to remove all the vowels from a given string.

Code:

```
s = input()
for i in s:
    if i not in "aeiouAEIOU":
        print(i, end="")
```

Input:

Hello World

Output:

Hll Wrld

14. Write a Python program to check if a given string contains only digits.

Code:

```
s = input()
if s.isdigit():
    print("Yes")
else:
    print("No")
```

Input:

12345

Output:

Yes

15. Write a Python program to find the most frequent word in a given sentence.

Code:

```
s = input()
words = s.split()
freq = {}
for word in words:
    if word in freq:
        freq[word] += 1
    else:
        freq[word] = 1
most_frequent = max(freq, key=freq.get)
print(most_frequent)
```

Input:

hello world hello python hello world

Output:

hello

16. Write a Python program to find the sum of digits of a given number using a while loop.

Code:

```
n = int(input())
sum = 0
while n > 0:
    sum = sum + n % 10
    n = n // 10
print(sum)
```

Input:

1234

Output:

10

17. Write a Python program to check if a given string is a palindrome or not using a for loop.

Code:

```
s = input()
reverse = ""
for i in s:
    reverse = i + reverse
if s == reverse:
    print("Palindrome")
else:
    print("Not Palindrome")
```

Input:

madam

Output:

Palindrome

18. Write a Python program to check if a number is a Lucky Number using recursive functions rSum(n) and rProd(n).

Code:

```
def rSum(n):
    if n == 0:
        return 0
    return n % 10 + rSum(n // 10)

def rProd(n):
    if n == 0:
        return 1
    return n % 10 * rProd(n // 10)

n = int(input())
s = rSum(n)
p = rProd(n)
if s == p:
```

```
        print("Yes")
    else:
        print("No")
```

Input:

1412

Output:

Yes

19. Write a Python program to find the area of a circle.

Code:

```
r = float(input())
area = 3.14 * r * r
print(area)
```

Input:

5

Output:

78.5

20. Write a Python program to check if a given number is a palindrome using recursion.

Code:

```
def reverse(n, rev=0):
    if n == 0:
        return rev
    return reverse(n // 10, rev * 10 + n % 10)

n = int(input())
if n == reverse(n):
    print("Palindrome")
else:
    print("Not Palindrome")
```

Input:

121

Output:

Palindrome